EE150 Signal and System
Homework 1

Note:

- Please provide enough calculation process to get full marks.
- Please submit your homework to Gradescope in PDF version.
- It's highly recommended to write every exercise on a single sheet of page.
- Late submissions will have points deducted according to the penalty policy.
- Please use English only to complete the assignment, solutions in Chinese are deducted according to the penalty policy.
- Plagiarizer will get zero points.
- The full score of this assignment is 100 points.

Exercise 1. (10pt)

Using Euler's formula $e^{j\theta} = \cos\theta + j\sin\theta$, derive the following identities:

(a) $\cos\theta = \frac{1}{2}\big(e^{j\theta} + e^{-j\theta}\big)$

(b) $\cos^2\theta = \frac{1}{2}\big(1 + \cos 2\theta\big)$

(c) $\sin\theta + \sin\phi = 2\sin\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right)$

Solution:

Exercise 2. (20pt)

Determine the energy E_∞ and power P_∞ of following signals. Which are finite-energy signals, which are finite-power signals, which are infinite energy and power signals? Write your calculation.

(a) $x(t) = e^{-t}u(4t)$

(b) $x(t) = e^{2jt}$

(c) $x[n] = \left(\frac{1}{3}\right)^n u[n]$

(d) $x[n] = 5\cos\left(\frac{\pi}{3}n\right)$

Solution:

Exercise 3. (15pt)

Determine whether or not each of the following signals is periodic. If the signal is periodic, determine its fundamental period.

(a) $x[n] = 2\cos(\frac{2\pi}{5}n) + \sin(\frac{4\pi}{7}n + \frac{\pi}{3})$

(b) $x[n] = 2\cos(\frac{2}{5}n) + \sin(\frac{4\pi}{7}n + \frac{\pi}{3})$

(c) $x(t) = 2\cos(\frac{2}{5}t) + \sin(\frac{4\pi}{7}t + \frac{\pi}{3})$

(d) $x[n] = \sin(\sqrt{2}\,\pi n)$

Solution:

Exercise 4. (10pt)

Consider the signals

$$x(t) = \cos \frac{2\pi t}{3} + 2 \sin \frac{16\pi t}{3},$$

$$y(t) = \sin \pi t.$$

Show that $z(t) = x(t)y(t)$ is periodic, and write $z(t)$ as a linear combination of harmonically related complex exponentials. That is, find a number T and complex numbers c_k such that

$$z(t) = \sum_k c_k e^{jk(2\pi/T)t}.$$

Solution:

Exercise 5.   (15pt)

In this chapter, we introduced a number of general properties of systems. In particular, a system may or may not be

(1) Memoryless

(2) Time invariant

(3) Linear

(4) Causal

(5) Stable

Determine which of these properties hold and which do not hold for each of the following continuous-time systems. Give your answers without explanation. In each example, $y(t)$ denotes the system output and $x(t)$ is the system input.

(a) $y(t) = x\left(\frac{t}{3}\right)$

(b) $y[n] = \sum_{k=-\infty}^{n} x[k]$,

(c) $y(t) = [\cos(3(t+1) - 2\pi)]x(t)$

Solution:

Exercise 6. (30pt)

(1) We have a signal x(t).

(a) Express x(t) in terms of the unit step function, then calculate $x'(t)$. (4 points)

(b) Plot the $x(-t+2)$ and $x(\frac{2}{3}t+1)$. (6 points)

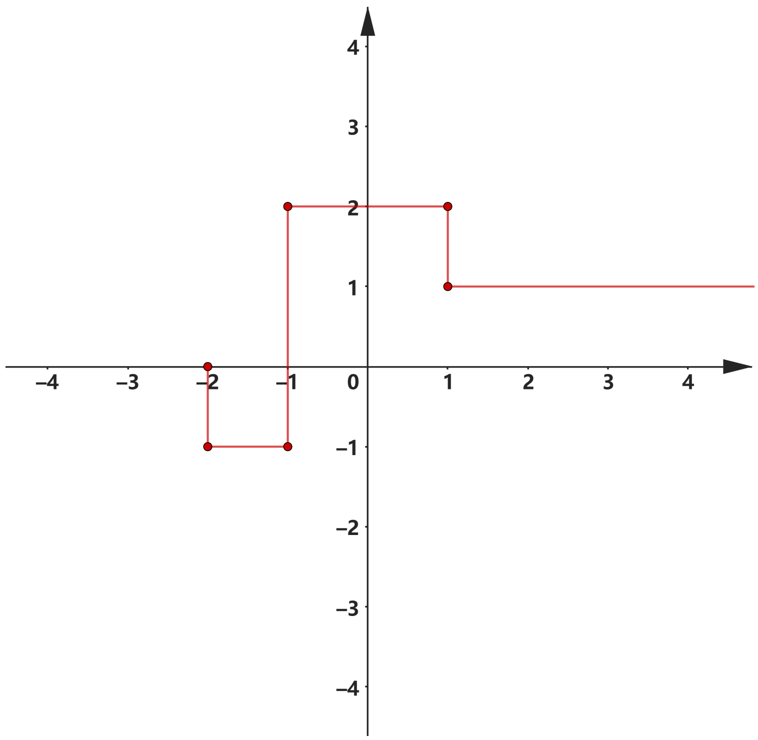

Figure 1: Signal x(t) in P6(1)

(2) We have a signal $x(t)$, the following figures are the parts of $x(t)$ and its odd part $x_o(t)$, for $t \geq 0$ only. Please plot the whole odd part $x_o(t)$, whole even part $x_e(t)$ and whole $x(t)$ for $-\infty < t < \infty$ and write the equation of each function. (Be careful to write the boundary values clearly)

$$x(t) = \begin{cases} 1+t, & 0 \leq t \leq 1, \\ 0, & t > 1, \end{cases} \qquad x_o(t) = \begin{cases} 2-t, & 0 \leq t \leq 2, \\ 0, & t > 2. \end{cases}$$

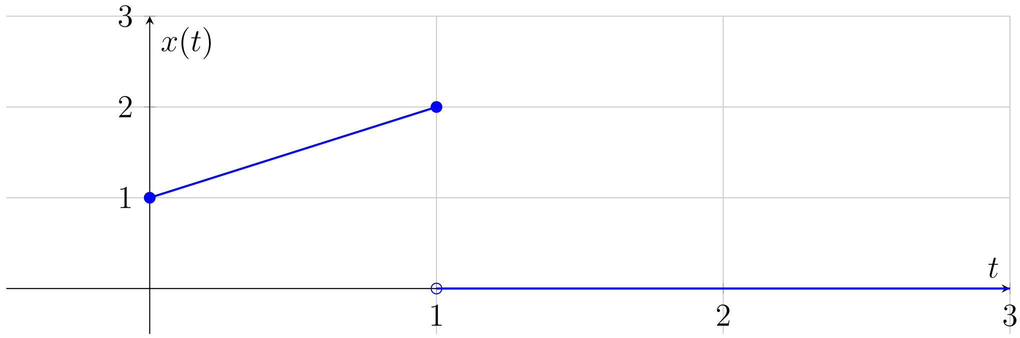

Figure 2: Original signal $x(t)$ for $t \geq 0$

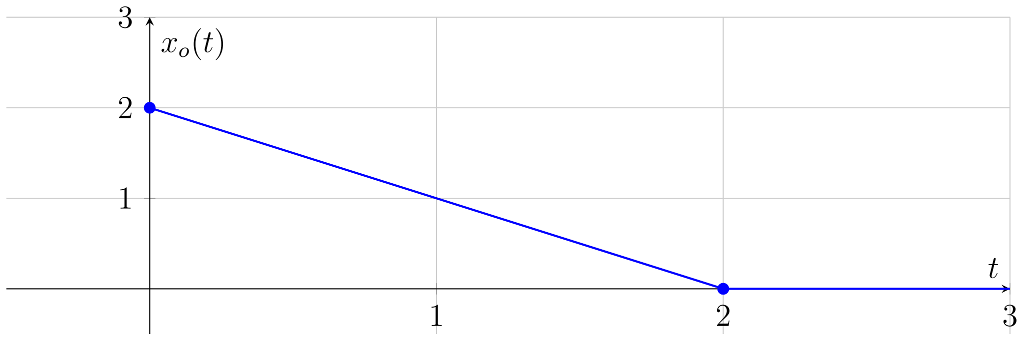

Figure 3: Odd part $x_o(t)$ for $t \geq 0$

Solution: